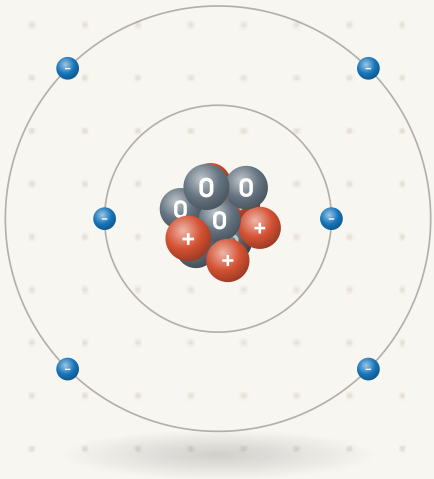
YEAR 11 PHYSICS, AQA GCSE, UNIT REVIEW

Atoms & Nuclear Radiation.

Unit review After Lesson 10

Mixed exam practice from the whole unit, a fix-it log, and extension pages for GCSE Physics (separate science) students.

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

REFERENCE

Toolkit.

Learn these: exam questions expect you to recall them without a data sheet.

The three particles

Lesson 1

Particle	Charge	Mass	Where
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	very small	energy levels

Reading the symbol

Lesson 1

$^{A}_{Z}X$ e.g. $^{23}_{11}Na$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A − Z; in an atom, electrons = Z
- Atom radius ≈ 1×10^{-10} m; nucleus radius < 1/10 000 of that

Isotopes and ions

Lesson 2

- **Isotopes**: same number of protons, different numbers of neutrons.
- **Positive ion**: an atom that has lost one or more outer electrons. The nucleus is unchanged.

Alpha scattering

Lesson 3

Observation	Conclusion
most straight through	mostly empty space
some deflected	positive centre
a very few bounced back	mass in a tiny centre

Models and energy levels

Lesson 4

- 1897 electron → 1904 plum pudding → 1911 nuclear → 1913 Bohr → 1932 neutron (Chadwick)
- **Absorb** EM radiation → higher level, further out.
- **Emit** EM radiation → lower level, closer in.

Nuclear radiation

Lesson 5

- Unstable nuclei decay at **random** to become more stable.
- **Activity**: decays per second, in becquerel (Bq). **Count rate**: decays recorded per second by a detector.
- α: 2p + 2n; β: fast electron, a neutron → proton; γ: EM radiation; n: a neutron

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Nuclear equations

Lesson 6

Emitted	Symbol	Mass no.	Atomic no.
alpha	$^{4}_{2}\mathrm{He}$	− 4	− 2
beta	$^{0}_{-1}\mathrm{e}$	no change	+ 1
gamma	γ	no change	no change
neutron	$^{1}_{0}\mathrm{n}$	− 1	no change

Top numbers balance; bottom numbers balance.

$$^{226}_{88}\mathrm{Ra} \rightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$$

$$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e}$$

Properties of radiation

Lesson 7

	Alpha	Beta	Gamma
Ionising	strongly	moderately	weakly
Range in air	a few cm	about 1 m	very far
Stopped by	paper, skin	a few mm of aluminium	reduced by thick lead or concrete
Typical use	smoke alarm	thickness gauge	medical tracer

More ionising → less penetrating.

Half-life from a graph

Lesson 8

1. Read the start value.
2. Halve it.
3. Across to the curve, down to the time axis.
4. Check with a second halving.

Half-life calculations

Lesson 9

- Number of half-lives = time ÷ half-life.
- After n half-lives, $(½)^n$ is left: ½, ¼, ⅛, 1/16…
- Net decline = initial − final; as a ratio of the initial, $1 - (½)^n$: ½, ¾, ⅞…

Contamination and irradiation

Lesson 10

- **Irradiation**: exposed to radiation; does not become radioactive.
- **Contamination**: radioactive atoms on or in an object.
- Outside the body γ is most hazardous; inside, α is.
- Tongs, distance, short time, lead, dosimeter.

AFTER LESSON 10, AT HOME OR IN CLASS

Unit review.

Mixed questions from the whole unit, like a real exam. Try each question without looking back. Then mark it with your teacher's answers and fill in the "fix it" log.

AQA-STYLE Mixed exam practice

42 marks, about 50 minutes

1 Complete the table. **[3 marks]**

Particle	Relative charge	Relative mass	Where in the atom?
proton	+1	1	nucleus
neutron	0	1	nucleus
electron	−1	very small	energy levels around the nucleus

2 An atom of cobalt is $^{59}_{27}\mathbf{Co}$. (a) Give the numbers of protons, neutrons and electrons. (b) The atom loses two electrons. How many electrons does the ion have, and what is its charge? **[3 marks]**

(a) 27 protons, 32 neutrons, 27 electrons (1). (b) 25 electrons (1); charge +2 (1).

3 Explain why $^{35}_{17}\mathbf{Cl}$ and $^{37}_{17}\mathbf{Cl}$ are isotopes of the same element. **[2 marks]**

Both have 17 protons, so they are the same element (1), but they have different numbers of neutrons, 18 and 20 (1).

4 Describe how the results of the alpha particle scattering experiment led to the nuclear model replacing the plum pudding model. **[6 marks]**

Level 3 (5–6): a clear account linking each result to a conclusion and explaining why the plum pudding model failed. Points: plum pudding: positive charge and mass spread out, so alpha particles expected to pass straight through with tiny deflections; most went straight through → mostly empty space; some deflected through large angles → positive centre repels positive alpha particles; a very few bounced back → mass and charge concentrated in a tiny centre; plum pudding could not explain large deflections/bounce-backs; so it was replaced by the nuclear model with a tiny, dense, positive nucleus and electrons outside.

5 Describe what happens to an electron in an atom when the atom emits electromagnetic radiation. **[2 marks]**

The electron moves to a lower energy level (1), closer to the nucleus (1).

6 Complete the equations using Th and Xe. The second shows beta decay. **[4 marks]**

$$^{238}_{92}U \rightarrow \boxed{^{234}_{90}Th} + {}^{4}_{2}He$$

$$^{131}_{53}I \rightarrow \boxed{^{131}_{54}Xe} + \boxed{^{0}_{-1}e}$$

Mass number 234 (1); atomic number 90 (1); 131 and 54 (1); beta particle $^{0}_{-1}e$ (1).

7 A student measures the count rate from a source with different absorbers. The background count rate is 25 counts per minute. Which types of radiation does the source emit? Explain your answer. **[3 marks]**

Absorber	none	paper	5 mm aluminium	2 cm lead
Count rate / counts per minute	410	405	150	60

No alpha: paper makes almost no difference (1). Beta: aluminium reduces the count a lot (1). Gamma: some gets through the aluminium and the lead, and the count is still above background (1).

8 A doctor needs a radioactive tracer that is injected into a patient and detected outside the body. Which isotope should be used? Give two reasons. **[3 marks]**

Isotope	Radiation emitted	Half-life
A	alpha	6 hours
B	gamma	6 hours
C	gamma	30 years
D	beta	8 days

B (1). Gamma passes out of the body to the detector and is only weakly ionising (1). A short half-life means it does not stay radioactive in the body for long, but lasts long enough for the scan (1).

9 The table shows the count rate from a sample. Determine the half-life. **[2 marks]**

Time / hours	0	1	2	3	4	5	6
Count rate / counts per second	640	453	320	226	160	113	80

640 halves to 320 at 2 hours (320 → 160 also takes 2 h) (1); half-life = 2 hours (1).

10 A source has an initial activity of 800 Bq. Calculate its activity after 4 half-lives, and the net decline as a ratio of the initial activity. **[2 marks]**

800 → 400 → 200 → 100 → 50 Bq (1); net decline 750 : 800 = 15 : 16 (1).

11 Explain why a sample of 50 unstable nuclei rarely has exactly 25 left after one half-life. **[2 marks]**

Radioactive decay is random (1); in a small sample the number that decay varies around half (1).

12 Compare the hazards of irradiation and contamination by an alpha source. **[4 marks]**

Irradiation by alpha from outside is low risk: alpha is stopped by the skin or a few cm of air (1), and the person does not become radioactive (1). Contamination: if the alpha source is breathed in or swallowed, it is inside the body (1), where its strongly ionising alpha particles damage nearby cells, and it keeps emitting until it decays or is removed (1).

13 Explain why scientists' model of the atom has changed over time. Use one example. **[3 marks]**

Models change when new evidence does not fit the old model (1). Example: alpha particles bouncing back could not be explained by the plum pudding model (1), so it was replaced by the nuclear model (1). (Or Bohr's energy levels / Chadwick's neutron.)

14 Explain why beta radiation is used to monitor the thickness of paper. **[3 marks]**

Alpha would be stopped completely by the paper; gamma would pass through almost unaffected (1). Beta is partly absorbed (1), so the count rate changes when the thickness changes (1).

REVIEW LATER

Fix-it log

Turn every lost mark into a target.

Q	What went wrong?	The correct physics (or method) is…	Lesson to revisit

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TRIPLE SCIENCE ONLY (AQA 4.4.3)

Background radiation and medical uses.

Only for students taking GCSE Physics (separate science). Combined Science students can skip these pages.

Background radiation

Background radiation is around us all the time. It comes from **natural sources**, such as rocks (for example granite, which gives off radon gas) and cosmic rays from space, and from **man-made sources**, such as fallout from nuclear weapons testing and nuclear accidents. The level depends on where you live and on your job.

The radiation dose a person receives is measured in **sieverts (Sv)**: 1000 millisieverts (mSv) = 1 Sv. When you measure a source, subtract the background count rate to get the corrected count rate.

Different half-lives, different hazards

Compare the same number of radioactive nuclei. An isotope with a **short half-life** has a high activity at first, but its activity falls quickly. One with a **long half-life** stays radioactive for a very long time, so waste containing it must be stored safely for thousands of years.

Nuclear radiation in medicine

Use	How it works	Radiation and half-life
Exploring (tracer)	a gamma emitter is injected or swallowed; a detector outside the body shows where it goes	gamma, short half-life (e.g. technetium-99m, 6 hours)
Treating cancer (radiotherapy)	beams of gamma rays from outside the body, or a source placed inside a tumour, kill cancer cells	gamma (or beta from an implant); beams are aimed to spare healthy cells

! Doctors weigh the **risk** (damage to healthy cells) against the **benefit** (finding or treating an illness).

PRACTISE

Check it

Answers are in the answer edition.

1 Name one natural and one man-made source of background radiation. [2 marks]

Natural: rocks / radon gas / cosmic rays (1). Man-made: fallout from nuclear weapons testing / nuclear accidents (1).

2 A detector records 185 counts per minute from a source. The background count rate is 25 counts per minute. Calculate the corrected count rate. **[1 mark]**

185 − 25 = 160 counts per minute.

3 Explain why a tracer injected into a patient should have a short half-life and emit gamma radiation. **[4 marks]**

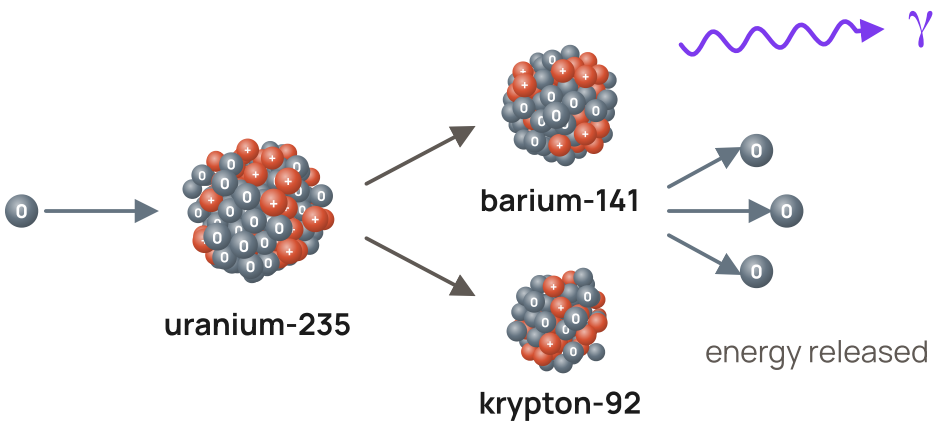
Gamma passes out of the body to be detected (1) and is only weakly ionising, so it causes little damage (1). A short half-life means the activity falls quickly, so the patient's dose is small (1), but it lasts long enough to complete the scan (1).

TRIPLE SCIENCE ONLY (AQA 4.4.4)

Nuclear fission and fusion.

Only for students taking GCSE Physics (separate science). These ideas build on nuclear equations (Lesson 6).

Nuclear fission



One possible fission of uranium-235. Top numbers: 235 + 1 = 141 + 92 + 3. Bottom numbers: 92 = 56 + 36.

Fission is the splitting of a large, unstable nucleus, such as uranium-235 or plutonium-239. Spontaneous fission is rare: usually the nucleus must first **absorb a neutron**. It then splits into **two smaller nuclei** of roughly equal size, and gives out **two or three neutrons** and gamma rays. **Energy** is released: all the products move away with kinetic energy.

The neutrons can go on to cause more fissions: a **chain reaction**. In a nuclear reactor it is **controlled** so energy is released steadily; in a nuclear weapon it is uncontrolled, causing an explosion.

Nuclear fusion

Fusion is the joining of two light nuclei, such as isotopes of hydrogen, to form a heavier nucleus. Some of the mass is converted into energy, which is given out as radiation. Fusion releases the energy of the Sun and other stars.

PRACTISE

Check it

Answers are in the answer edition.

1 Draw a diagram to show how one fission can start a chain reaction. Label the neutrons and the new nuclei. **[3 marks]**

A neutron hits a U-235 nucleus (1); it splits into two smaller nuclei and 2 or 3 neutrons (1); each of those neutrons hits another U-235 nucleus, which also splits (1).

2 Compare nuclear fission with nuclear fusion. **[4 marks]**

Fission splits a large nucleus; fusion joins two light nuclei (1). Fission usually needs a neutron to be absorbed first; fusion joins nuclei such as hydrogen isotopes (1). Both release energy (1). Fission is used in nuclear power stations; fusion powers the stars (1).

CHECKPOINT

Quick check

Slide 55, Triple Science, using the nucleus. Answer each question, then check it as your teacher reveals the answer.

1 Name two natural sources of background radiation.

Rocks (radon gas) and cosmic rays.

2 Why do aircrew get a higher dose?

There are more cosmic rays at high altitude.

3 Why does a medical tracer need a short half-life?

It lasts for the scan, then soon decays away.

4 What must a uranium nucleus absorb before fission?

A neutron.

5 What do control rods do?

They absorb neutrons to control the chain reaction.

6 Fusion joins two light nuclei. Where does the energy come from?

Some of the mass is converted into energy.

REFERENCE

Key words.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

Absorb: Take in energy. An electron that absorbs EM radiation moves to a higher energy level.

Activity: The rate at which a source of unstable nuclei decays. Unit: becquerel (Bq).

Alpha particle (α): Two protons and two neutrons: a helium nucleus. Strongly ionising; stopped by paper.

Alpha scattering: The experiment in which alpha particles fired at thin gold foil showed that atoms have a tiny, positive nucleus.

Atom: The smallest part of an element. It has a tiny nucleus with electrons around it and no overall charge.

Atomic number: The number of protons in the nucleus. It decides the element.

Becquerel (Bq): Unit of activity: 1 Bq = 1 decay per second.

Beta particle (β): A high-speed electron ejected from the nucleus when a neutron turns into a proton.

Bohr model: Electrons orbit the nucleus at specific distances, called energy levels (1913).

Contamination: Unwanted radioactive atoms on or in an object.

Count rate: The number of decays recorded each second by a detector.

Daughter nucleus: The new nucleus formed when a nucleus decays.

Dosimeter: A badge that measures how much radiation a worker receives.

Electron: A tiny particle with a relative charge of −1, found in energy levels around the nucleus.

Emit: Give out. An electron that moves to a lower energy level emits EM radiation.

Energy level: A specific distance from the nucleus at which electrons are found.

Evidence: Observations and measurements used to test a model.

Gamma ray (γ): Electromagnetic radiation emitted from a nucleus. Weakly ionising; very penetrating.

Half-life: The time for the number of unstable nuclei in a sample, or the count rate, to halve.

Ion: An atom that has lost (or gained) electrons, so it has an overall charge.

Ionising: Able to knock electrons off atoms, turning them into ions.

Irradiation: Exposing an object to nuclear radiation. The object does not become radioactive.

Isotopes: Atoms of the same element with different numbers of neutrons.

Mass number: The total number of protons and neutrons in the nucleus.

Model: A simplified explanation of something we cannot see directly; changed when new evidence does not fit.

Net decline: How much the activity has fallen; often given as a ratio of the initial activity.

Neutron: A particle in the nucleus with no charge and a relative mass of 1.

Nuclear equation: An equation showing a decay, in which the mass numbers and atomic numbers balance.

Nuclear model: A tiny, dense, positive nucleus with electrons outside it (Rutherford, 1911).

Nucleus: The tiny centre of an atom, containing protons and neutrons and almost all the mass.

Peer review: Checking of published findings by other scientists.

Penetration: How far radiation can travel through materials before it is absorbed.

Plum pudding model: A ball of positive charge with negative electrons embedded in it (1904).

Proton: A particle in the nucleus with a relative charge of +1 and a relative mass of 1.

Radioactive decay: An unstable nucleus giving out radiation to become more stable. It is random.

Random: Impossible to predict for one nucleus: which one decays next, or when.

Unstable nucleus: A nucleus that will decay and give out radiation.

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FINISH

Progress tracker.

Colour one circle for each statement: **red** = not yet, **amber** = getting there, **green** = I can do this without help. Revisit your reds first.

I CAN…	R / A / G
01: Inside the atom	
describe the charge, mass and position of each particle	
state the size of an atom and of its nucleus	
explain why an atom is neutral	
work out protons, neutrons and electrons from nuclear notation	
02: Isotopes and ions	
explain what isotopes are	
use nuclear notation to compare isotopes	
describe how a positive ion forms	
work out the charge on an ion	
03: From plum pudding to the nucleus	
describe the plum pudding model	
describe the alpha scattering experiment	
link each result to a conclusion about the atom	
compare the plum pudding and nuclear models	
04: Energy levels and the nucleus	
describe Bohr's model of the atom	
explain what happens when electrons absorb or emit EM radiation	
describe the discovery of protons and neutrons	
explain why models of the atom have changed	
05: Radioactive decay	
explain why unstable nuclei decay	
describe decay as random	
define activity and count rate	
describe alpha, beta, gamma and neutron radiation	

I CAN…	R / A / G
06: Nuclear equations	
write the symbols for alpha, beta, gamma and neutron radiation	
balance a nuclear equation	
find the new nucleus after alpha or beta decay	
explain the effect of each decay on mass and charge	
07: Properties and uses of radiation	
compare the properties of alpha, beta and gamma	
explain the link between ionising and penetration	
identify radiation from absorber data	
choose and justify a source for a use	
08: Half-life and random decay	
explain why decay is random but predictable	
define half-life in two ways	
find a half-life from a decay curve	
plot a decay curve from data	
09: Half-life calculations	
calculate how much is left after a number of half-lives	
find a half-life from data or a graph	
find how long a source takes to fall to a value	
express the net decline as a ratio	
10: Contamination and irradiation	
explain the difference between irradiation and contamination	
compare the hazards inside and outside the body	
describe precautions and how each reduces the risk	
explain why findings are published and peer reviewed	

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